

LECTURE 5

MINING WEB

CONTENT II

LEK HSIANG HUI

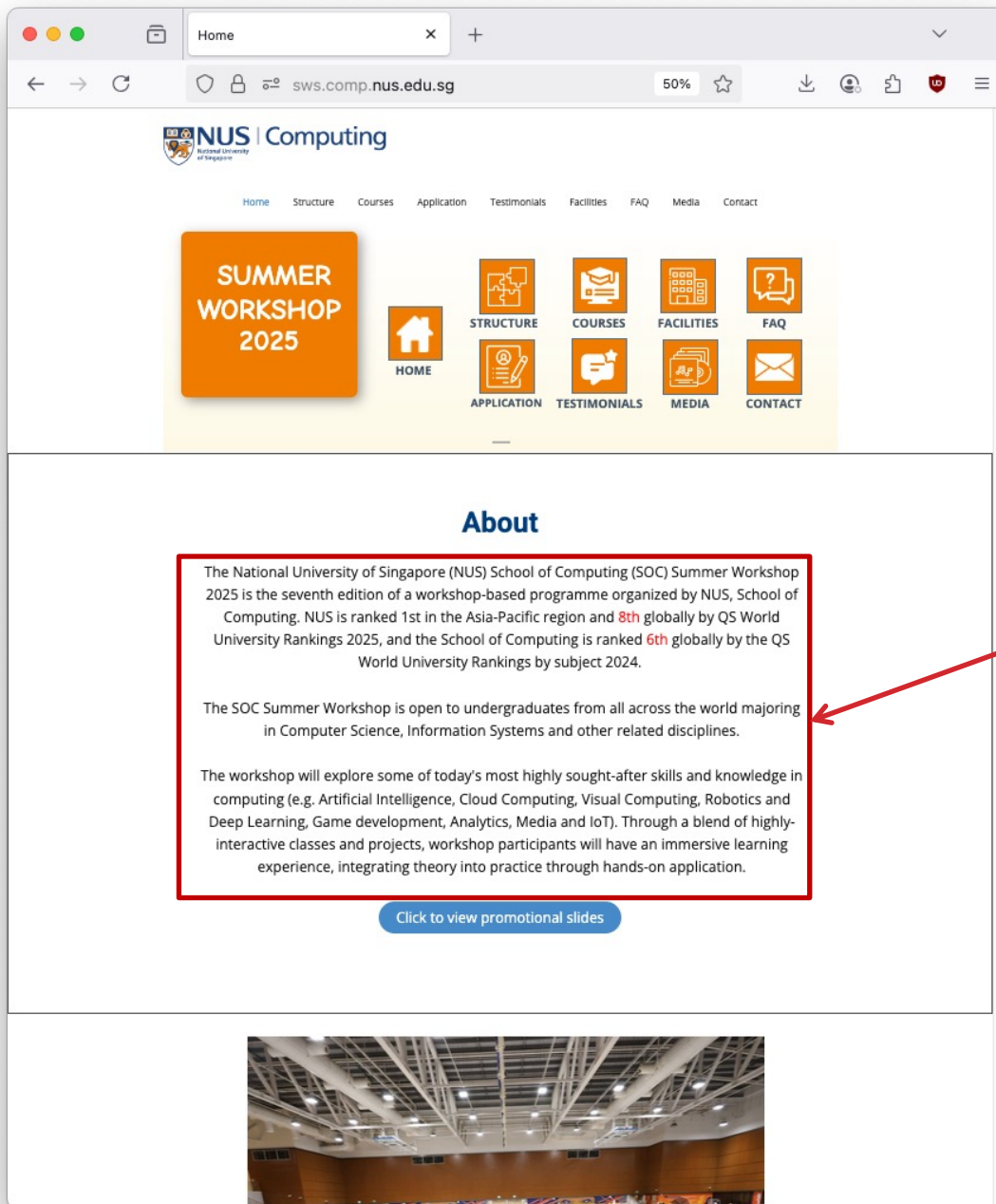
OUTLINE

Document Object Model (DOM)

XPath

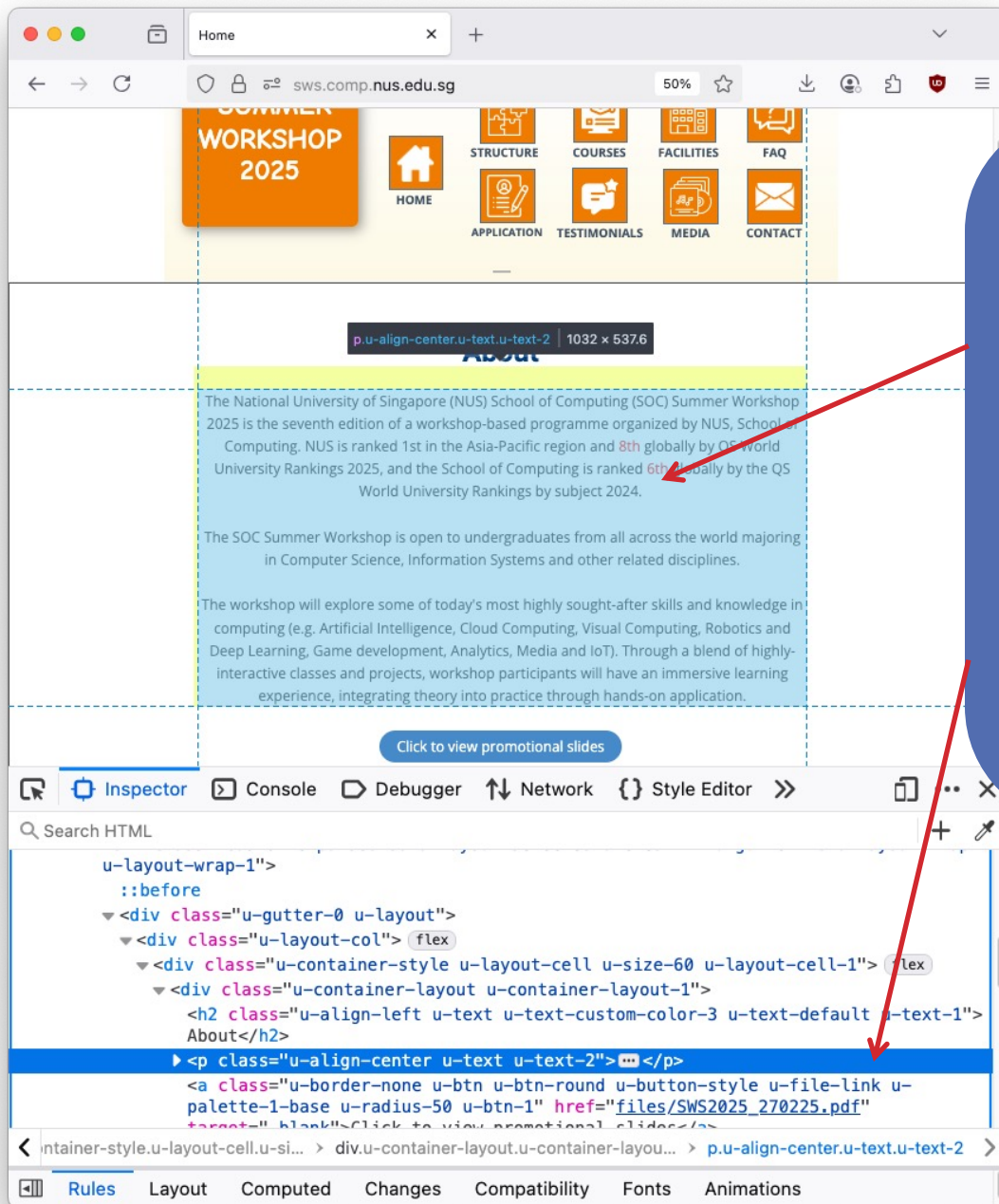
CSS Selectors

Extracting Content using HTML Parser



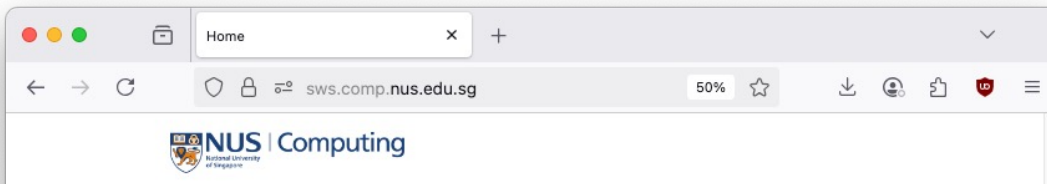
Recap: Extracting contents from HTML source

Suppose we want to extract the contents under the About



First figure out the rough position in the page using the web browser's Inspect Element feature

Then find it View Source window (because the browser might clean up the page slightly such as removing unnecessary spaces)



Write the regular expression:

```
<p class="u-align-center u-text u-text-2">(.*)</p>
```

About

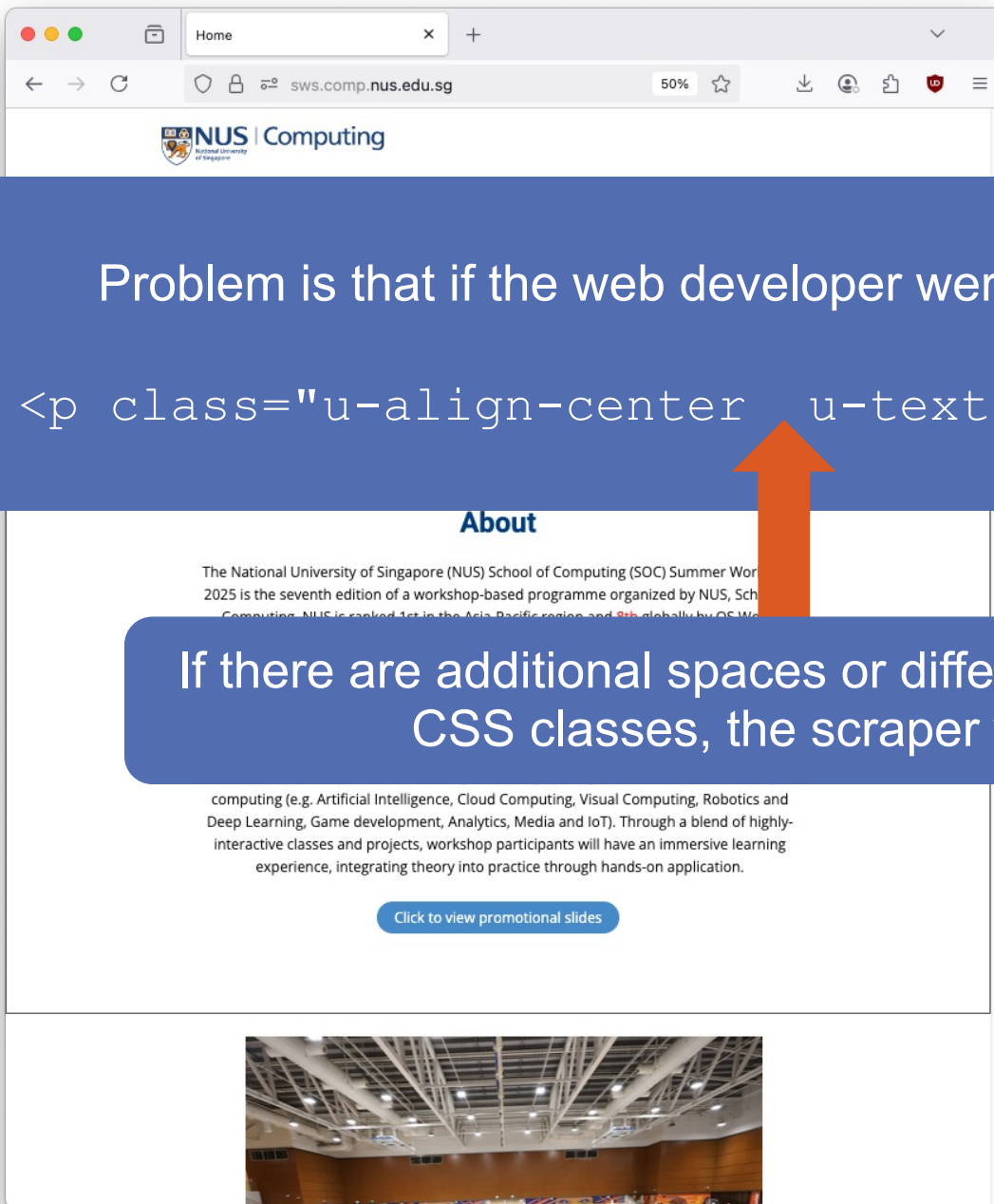
The National University of Singapore (NUS) School of Computing (SOC) Summer Workshop 2025 is the seventh edition of a workshop-based programme organized by NUS, School of Computing. NUS is ranked 1st in the Asia-Pacific region and 8th globally by QS World University Rankings 2025, and the School of Computing is ranked 6th globally by the QS World University Rankings by subject 2024.

The SOC Summer Workshop is open to undergraduates from all across the world majoring in Computer Science, Information Systems and other related disciplines.

The workshop will explore some of today's most highly sought-after skills and knowledge in computing (e.g. Artificial Intelligence, Cloud Computing, Visual Computing, Robotics and Deep Learning, Game development, Analytics, Media and IoT). Through a blend of highly-interactive classes and projects, workshop participants will have an immersive learning experience, integrating theory into practice through hands-on application.

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Problem is that if the web developer were to write the HTML as:

```
<p class="u-align-center u-text u-text-2">(.*?)</p>
```

If there are additional spaces or different ordering for the CSS classes, the scraper will break

FLAW OF STRING-BASED APPROACH OF WEB SCRAPING

Too easily affected by the way how the HTML is written

- Minor changes (e.g. newlines, spaces, capitalization, shifting of attributes ordering, etc) might break the scraper
- Even when the page is still a totally valid page and might look exactly the same

RECAP: TECHNIQUES FOR WEB SCRAPING

The following are some of the techniques for doing web scraping:

- Extracting content from HTML source
- **Extracting content using a HTML parser**
- Web Scraping using APIs
- Scraping using an actual browser/headless browser

We will look at another approach which is more robust against this situation

DOCUMENT OBJECT MODEL (DOM)

Document
Object Model
(DOM)

XPath

CSS
Selectors

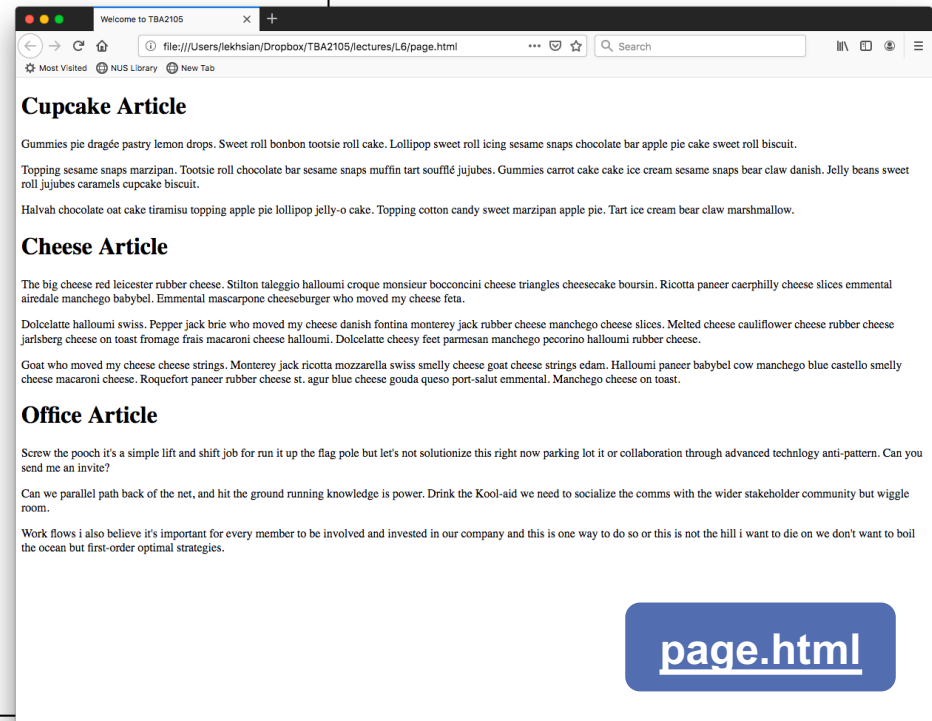
Extracting
Content using
HTML Parser

```

<html>
  <head>
    <meta charset="UTF-8" />
    <meta name="description" content="..." />
    <meta http-equiv="X-UA-Compatible" content="ie=edge" />
    <title>Welcome to SWS3023</title>
  </head>
  <body>
    <div class="article" id="a0042">
      <h1>Cupcake Article</h1>
      <div class="header">...</div>
      <p>...</p>
      <p>...</p>
    </div>
    <div class="article" id="a0043">
      <h1>Cheese Article</h1>
      <div class="header">...</div>
      <p>...</p>
      <p>...</p>
    </div>
    <div class="article special" id="b0051">
      <h1>Office Article</h1>
      <div class="snippet">...</div>
      <p>...</p>
      <p>...</p>
    </div>
  </body>
</html>

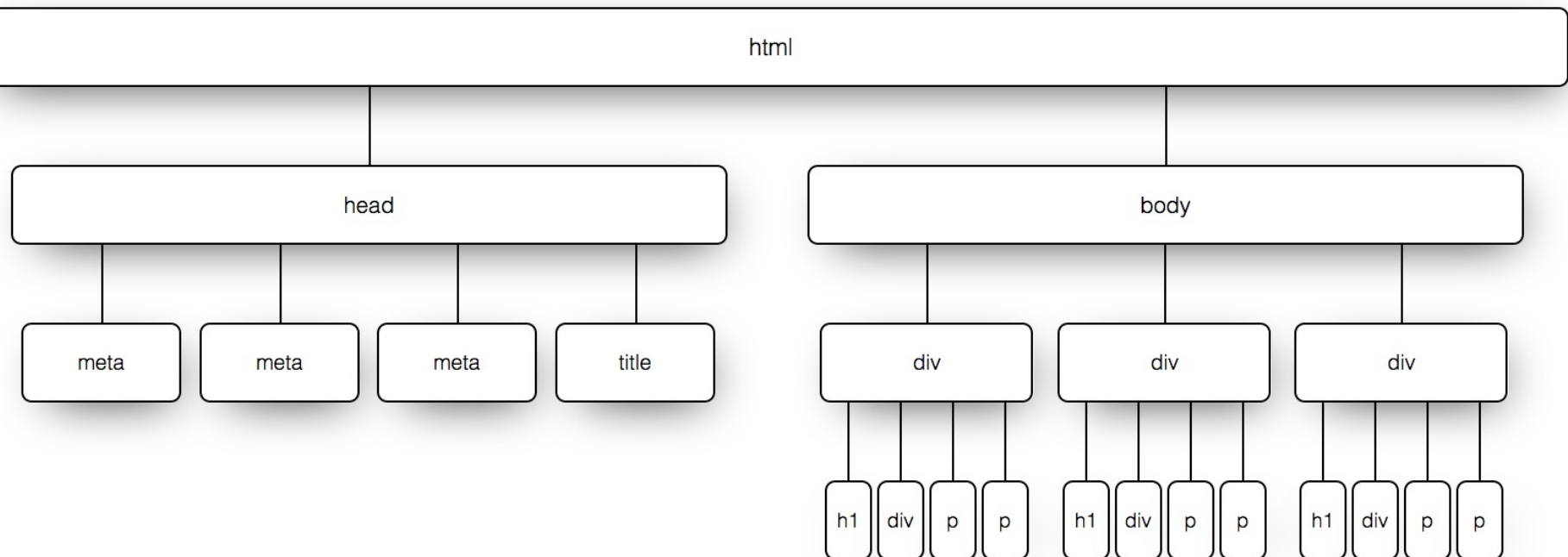
```

Recall: HTML defines
the content and the
layout of the page



DOCUMENT OBJECT MODEL (DOM)

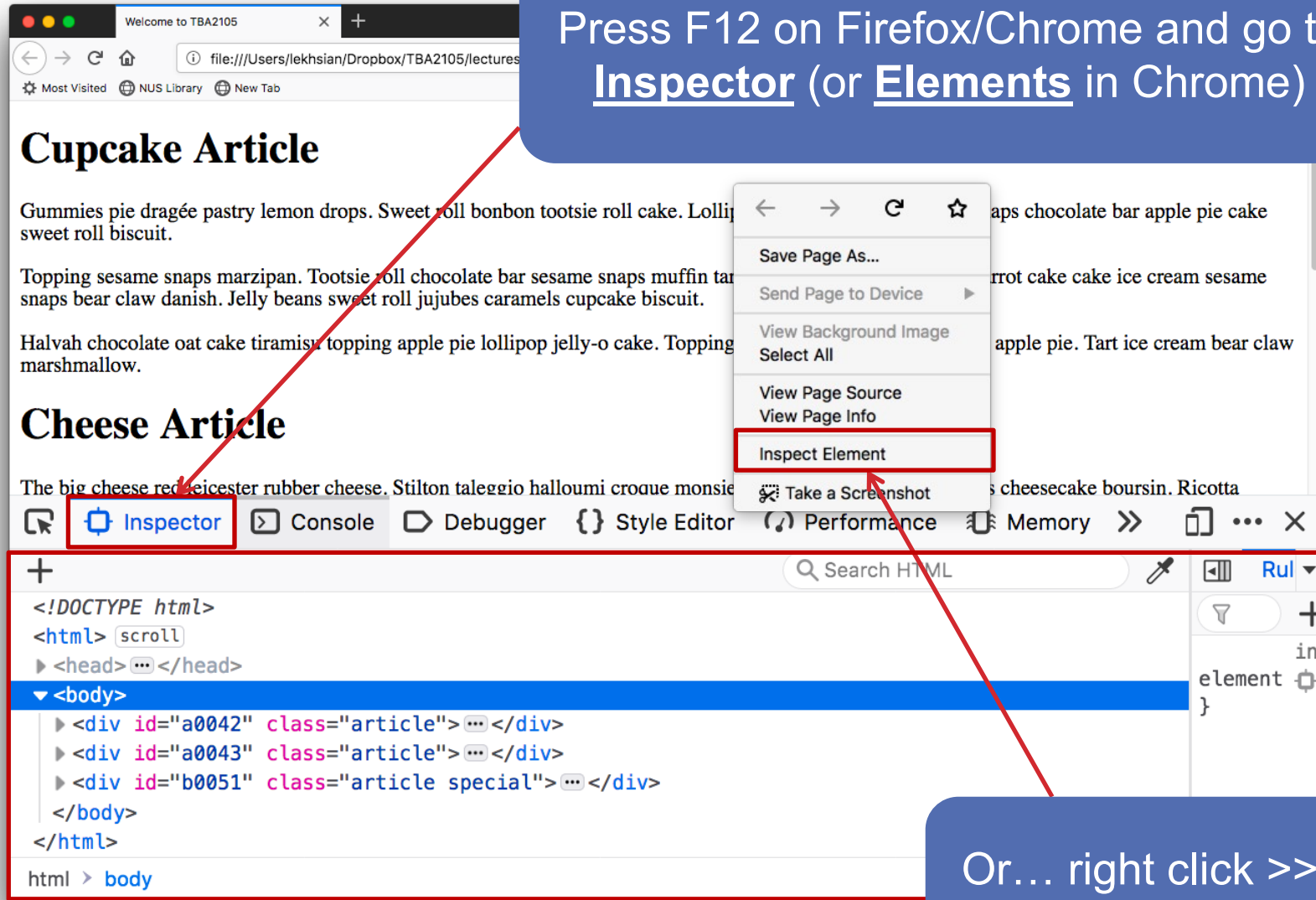
When the browser loads a page, it creates a **DOM** of the page and use it to render the page



DOM tree of page.html

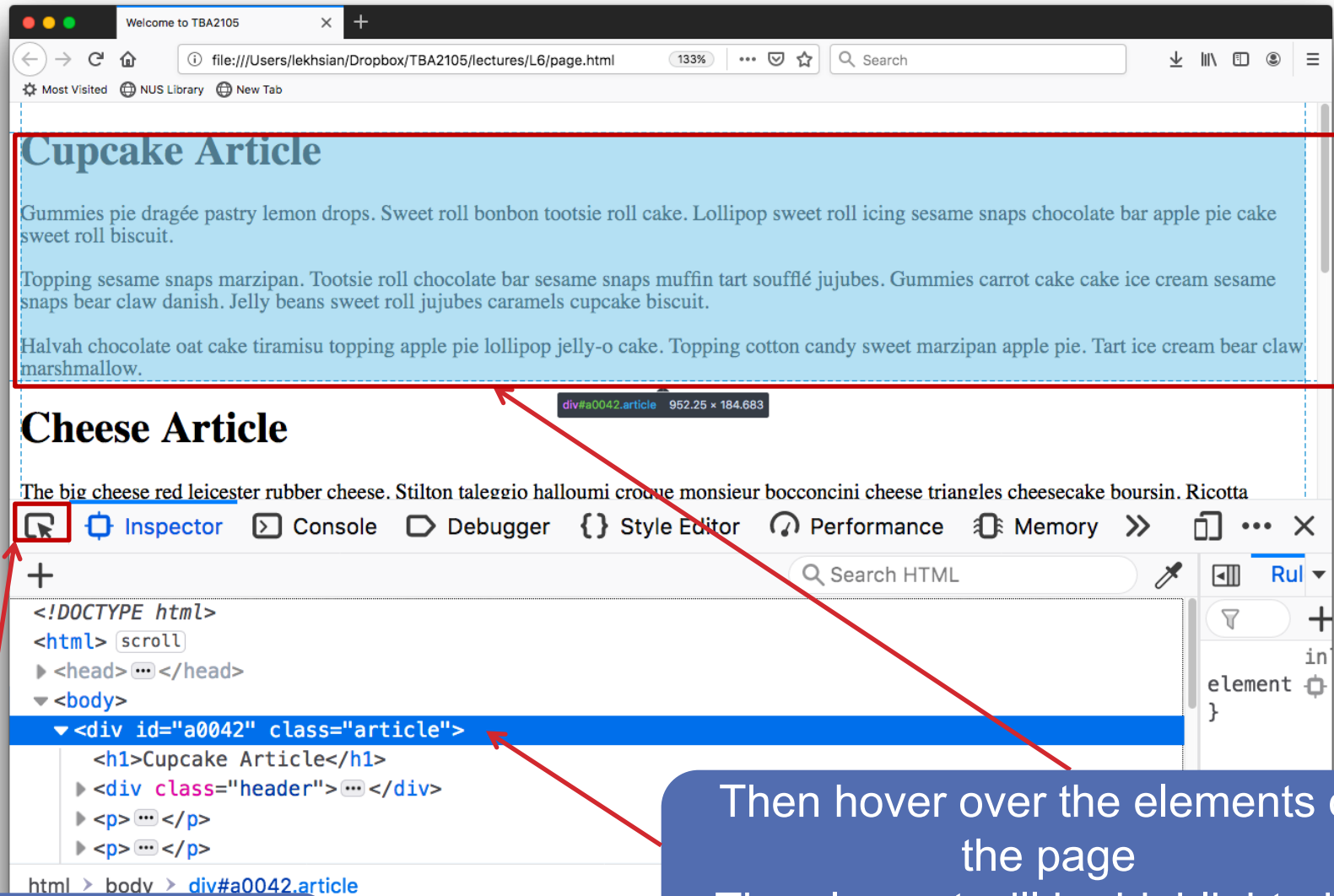
VIEWING THE DOM ON WEB BROWSER

Press F12 on Firefox/Chrome and go to the Inspector (or Elements in Chrome) tab



Or... right click >>
Inspect Element

VIEWING THE DOM ON WEB BROWSER



First Choose the
Pick Element Icon

Then hover over the elements on
the page
The element will be highlighted at
the Inspector tab

VIEWING THE DOM ON WEB BROWSER

Welcome to TBA2105

file:///Users/lekhsian/Dropbox/TBA2105/lectures/L6/page.html 133%

Cupcake Article

Gummies pie dragée pastry lemon drops. Sweet roll bonbon tootsie roll cake. Lollipop sweet roll icing sesame snaps chocolate bar apple pie cake sweet roll biscuit.

Topping sesame snaps marzipan. Tootsie roll chocolate bar sesame snaps muffin tart soufflé jujubes. Gummies carrot cake cake ice cream sesame snaps bear claw danish. Jelly beans sweet roll jujubes caramels cupcake biscuit.

Halvah chocolate oat cake tiramisu topping apple pie lollipop jelly-o cake. Topping cotton candy sweet marzipan apple pie. Tart ice cream bear claw marshmallow.

Cheese Article

The big cheese red leicester rubber cheese. Stilton taleggio halloumi croque monsieur bocconcini cheese triangles cheesecake boursin. Ricotta

Inspector Console Debugger Style Editor Performance Memory

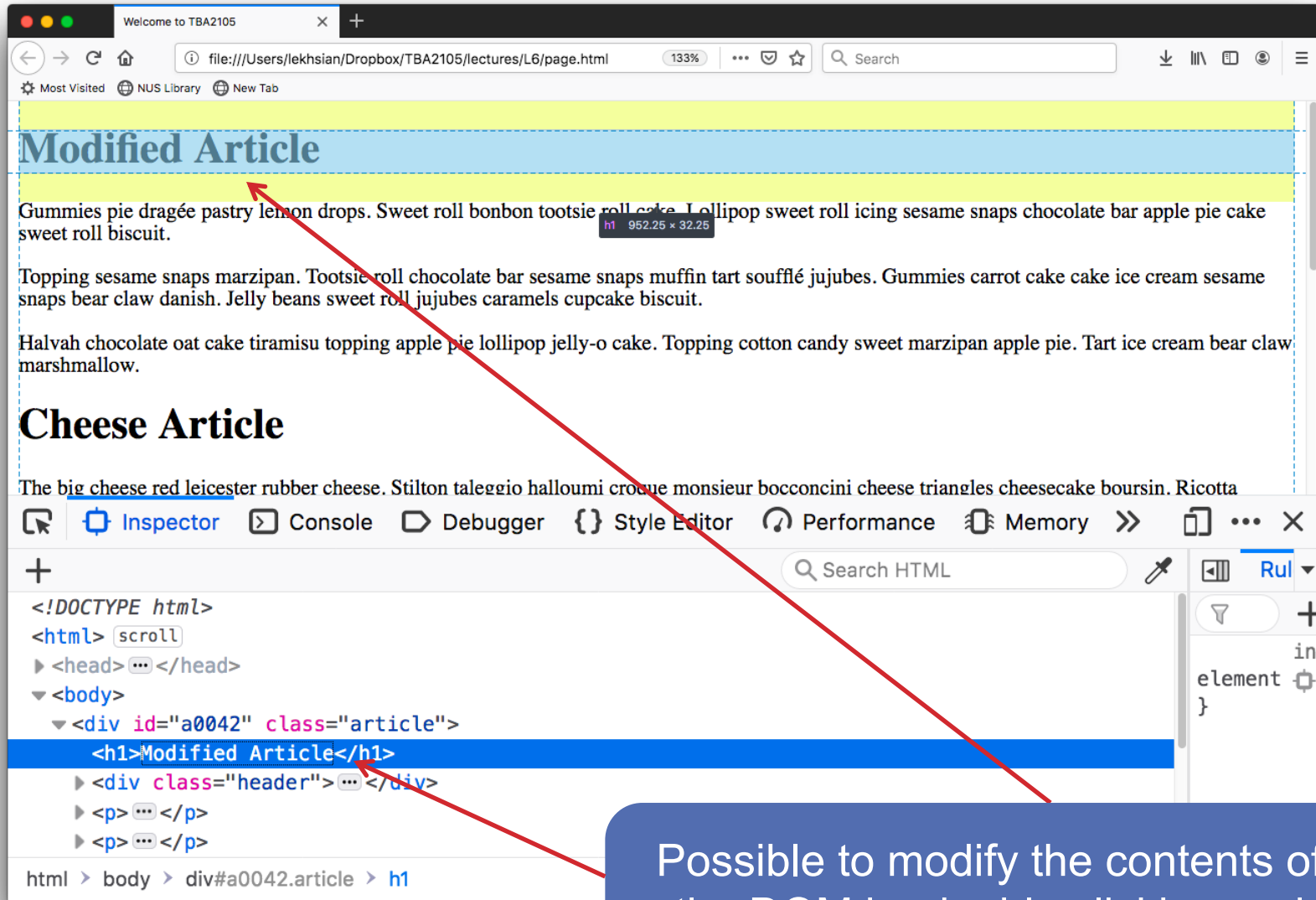
Search HTML

```
<!DOCTYPE html>
<html> scroll
  <head> ... </head>
  <body>
    <div id="a0042" class="article">
      <h1>Cupcake Article</h1>
      <div class="header"> ... </div>
      <p> ... </p>
      <p> ... </p>
    </div>
  </body>
</html>
```

html > body > div#a0042.article

Alternatively, could just select an element at the Inspector tab, the selected element will be highlighted on the page

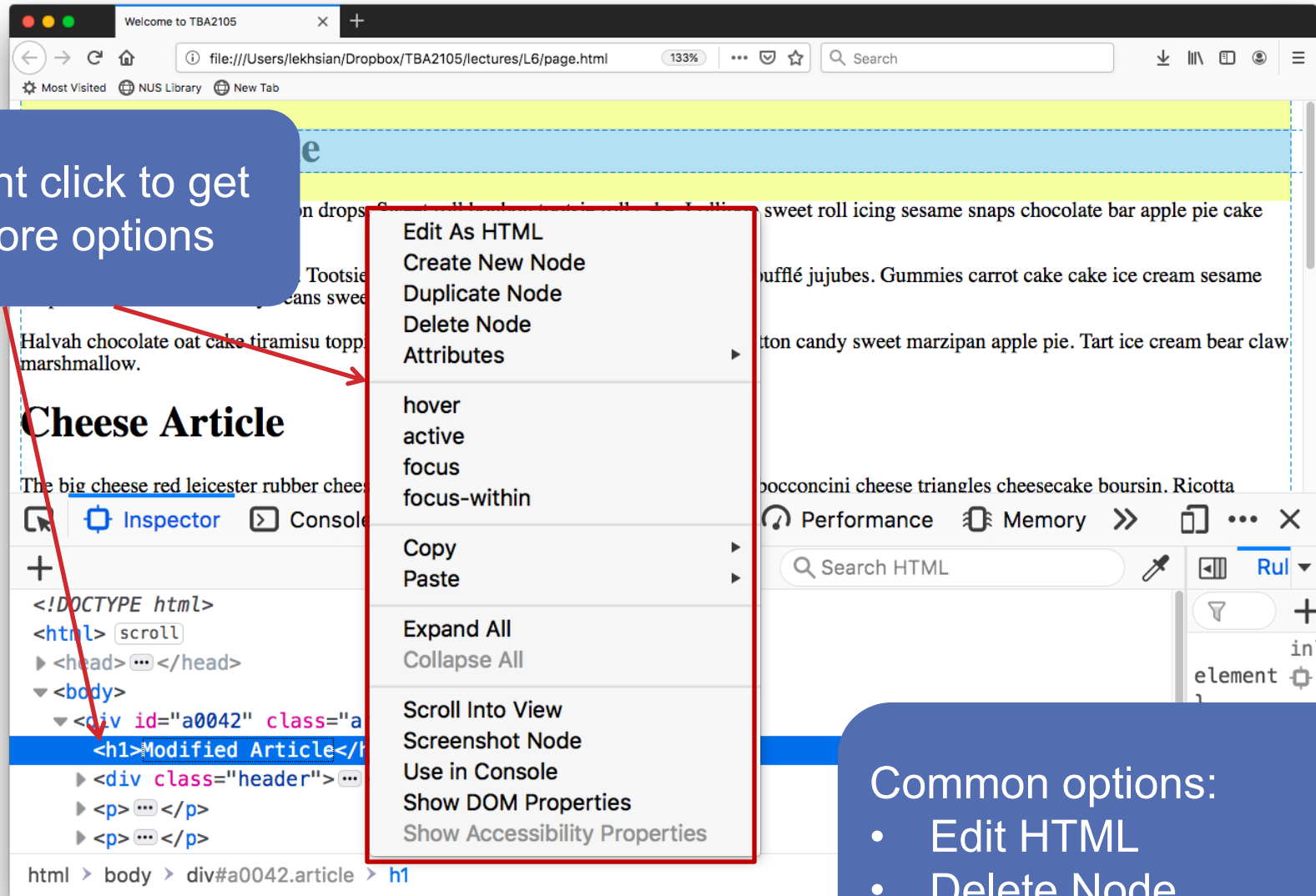
MODIFYING THE DOM ON WEB BROWSER



Possible to modify the contents of the DOM by double clicking and editing

MODIFYING THE DOM ON WEB BROWSER

Right click to get more options



Common options:

- Edit HTML
- Delete Node
- Copy (to be elaborated later)

NAVIGATING THE DOM TREE

Possible to navigate the DOM programmatically:

- e.g. extract content, add/remove elements, etc

2 ways to navigate the DOM tree

- Using **XPath**
- Using **CSS Selectors**

XPATH

Document
Object Model
(DOM)

XPath

CSS
Selectors

Extracting
Content using
HTML Parser

XPATH

XPath is a way to navigate through elements and attributes in an **eXtensible Markup Language (XML)** document

- HTML is very similar to a XML document
- Thus, XPath can be used to navigate through HTML documents also

XPATH EXPRESSIONS

XPath Expression	Description
/html	Select the html node
/html/head	Select the head node (notice the navigation path)
/html/body/div	Select all the div <u>directly</u> under body node
/html/body/div[1]	Select the <u>first</u> div <u>directly</u> under body node
//div	Select all div in the document (regardless of its ancestor)
//div[1]	Select <u>all the first</u> div in the document (originating from an ancestor). Note that this can select <u>multiple div</u>
//div[last()]	Select <u>all the last</u> div in the document (originating from an ancestor). Note that this can select <u>multiple div</u>
//body/div	Select all the div nodes that is a direct child of the body node
//body//div	Select all the div nodes that is a descendent of the body node
//body/*	Select all the nodes under the body node (* is a wildcard)

TRYING XPATH ON THE BROWSER

1. Select the Console tab

3. Result of the node selection is return as an array (hovering over a value will highlight the element on the page)

2. Type `$x (XPATH_EXPRESSION)` (this is effectively JavaScript)

Modified Article

Gummies pie dragée pastry lemon drops. Sweet roll bonbon tootsie roll cake. Lollipop sweet roll icing sesame snaps chocolate bar apple pie cake sweet roll biscuit.

Topping sesame snaps marzipan. Tootsie roll chocolate bar sesame snaps muffin tart soufflé jujubes. Gummies carrot cake cake ice cream sesame snaps bear claw danish. Jelly beans sweet roll jujubes caramels cupcake biscuit.

Halvah chocolate oat cake tiramisu topping apple pie lollipop marshmallow.

Choose Article

```
>> $x('/html')
< ▶ Array [ html ]
>> $x('/html/head')
< ▶ Array [ head ]
>> $x('/html/body')
< ▶ Array [ body ]
>>
```

XPATH EXPRESSIONS

Exercise: Try out these expression using the browser **Console**

XPath Expression	Description
/html	Select the html node
/html/head	Select the head node (notice the navigation path)
/html/body/div	Select all the div <u>directly</u> under body node
/html/body/div[1]	Select the <u>first</u> div <u>directly</u> under body node
//div	Select all div in the document (regardless of its ancestor)
//div[1]	Select <u>all the first</u> div in the document (originating from an ancestor). Note that this can select <u>multiple div</u>
//div[last()]	Select <u>all the last</u> div in the document (originating from an ancestor). Note that this can select <u>multiple div</u>
//body/div	Select all the div nodes that is a direct child of the body node
//body//div	Select all the div nodes that is a descendent of the body node
//body//*	Select all the nodes under the body node (* is a wildcard)

XPATH EXPRESSIONS

XPath Expression	Description
<code>//div[@id]</code>	Select all div node with id attribute
<code>//div[@class='article']</code>	Select all div node with class attribute with the value article (must be exact match , i.e. even with extra spaces will not work)
<code>//*[@class='article']</code>	Select any node with class attribute with the value article
<code>//div[starts-with(@id, 'a004')]</code>	Select all div node with id attribute having value starting with a004
<code>//div[contains(@id, '00')]</code>	Select all div node with id attribute having value containing 00
<code>//*[contains(@class, 'article')]/..</code>	Select parent node of any node with class attribute having value containing article
<code>//div //p</code>	Select all div and p nodes

CASE SENSITIVITY

XPath Expression	Case Sensitive?
<code>//div[@id]</code> <code>//DIV[@id]</code> <code>//div[@ID]</code> <code>//DiV[@ID]</code>	All these are equivalent. Both the node name and the attribute names are not case sensitive
<code>//div[@class='article']</code> <code>//div[@class='Article']</code>	These are not equivalent. The attribute value is case sensitive Same principle applies to <code>starts-with()</code> and <code>contains()</code>

XPATH REFERENCES

https://www.w3schools.com/xml/xpath_syntax.asp

CSS SELECTORS



CSS SELECTORS

CSS selectors is another way to select elements in HTML

- Used mainly for selecting HTML elements in order to apply styling into the webpage
- But could also be used for referencing to elements in JavaScript or for doing web scraping
- More commonly used compared to XPath
- Tends to be shorter compared to XPath

CSS SELECTORS

CSS selectors is another way to select elements in HTML

- ...
- For the purpose of web scraping, these are the most useful type of selectors
 - **Element** selector
 - **Class** selector
 - **Id** selector
 - **Attribute** selector
 - **Pseudo-Classes** selector
 - **Relationship** selector

ELEMENT, CLASS, ID SELECTOR

Element selector used to select all elements with a certain type of tag

Id selector used to select the element with a certain id value

- HTML usually use `id` to uniquely identify an element

Class selector used to select the element with a certain class attribute value

- HTML elements usually use `class` to denote the style of the element

TRYING CSS SELECTORS

To use css selector, type
\$\$ (CSS_SELECTOR)

Element selector:

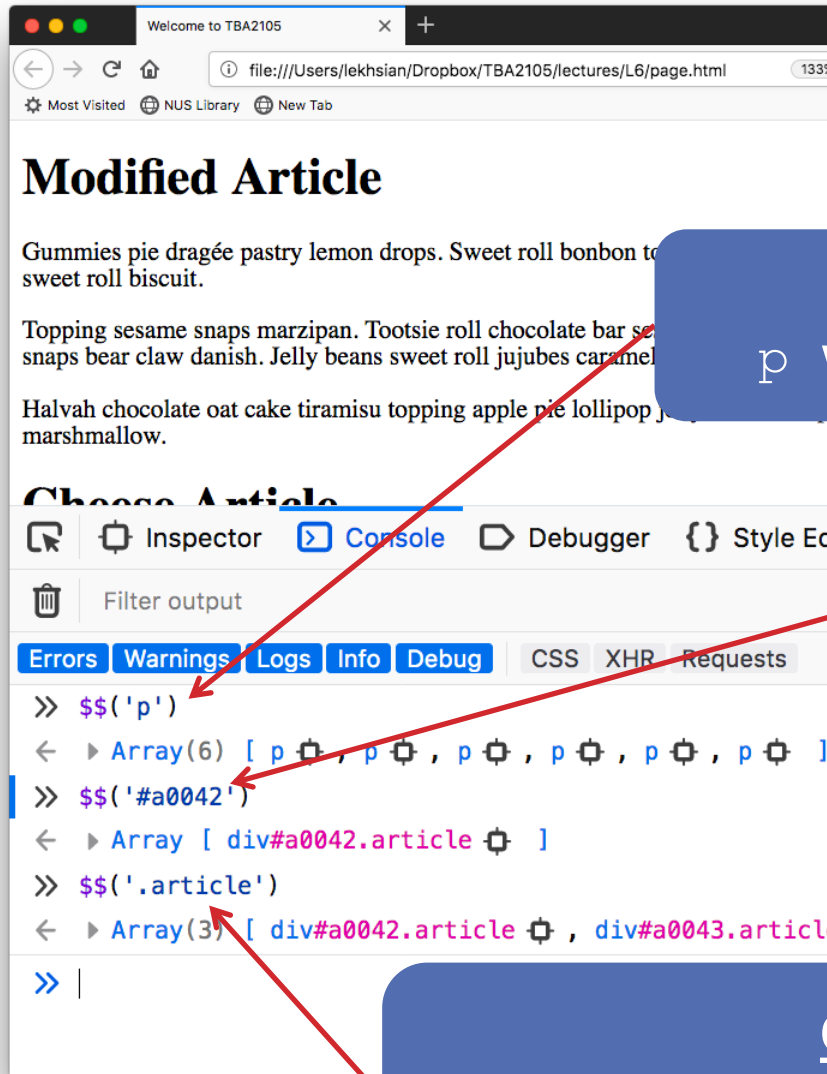
p will select all paragraph elements

Id selector:

#a0042 will select the element
with id = a0042

Class selector:

.article will select the elements with class
containing article class



CSS SELECTORS

Possible to have multiple type of selectors

CSS Selector	Description
<code>div#a0042</code>	This selects the div node with id = a0042
<code>div.article</code>	This selects all the div nodes with class <u>containing article</u> (need not be an exact match, the element can have multiple class values separated by spaces)
<code>.article#a0042</code>	This selects all nodes with class <u>containing article</u> and with id = a0042
<code>.article.special</code>	This selects all nodes with class containing both article and special (can be any ordering but be careful that <u>no spaces between the 2 classes</u>)
<code>div,p</code> <code>div , p</code> <code>div, p</code>	This selects all the div and p nodes

ATTRIBUTE SELECTORS

CSS Selector	Description
<code>meta[name]</code>	This selects the meta node with a name attribute
<code>meta[name='description']</code>	This selects the meta node with a name attribute with the value description (exact match)
<code>meta[content~='page']</code>	This selects the meta node with a content attribute containing a page as a <u>whole word match</u> . Would not match somepage.
<code>meta[content*='learn']</code>	This selects the meta node with a content attribute containing learn . Matches as long as there is a substring of learn.
<code>[class = 'article']</code> <code>*[class = 'article']</code>	This selects all node with class attribute that is article (exact match)
<code>[id ^= 'a00']</code>	This selects all node with id attribute that <u>starts with</u> a00
<code>[id \$= '51']</code>	This selects all node with id attribute that <u>ends with</u> a00

PSEUDO-CLASSES SELECTORS

CSS Selector	Description
<code>div:first-child</code>	This selects all div nodes that is the <u>1st child of its parent</u>
<code>p:last-child</code>	This selects all p nodes that is the <u>last child of its parent</u>
<code>p:nth-child(3)</code>	This selects all p nodes that is the <u>3rd node of its parent</u>

RELATIONSHIP SELECTORS

CSS Selector	Description
<code>body div</code>	This selects all div nodes that are <u>descendents</u> of body
<code>body > div</code>	This selects all div nodes that are <u>direct child</u> of body
<code>h1 + *</code>	This selects all nodes that is a <u>next sibling</u> of h1
<code>.article .snippet</code>	This will find all nodes having class containing article , and select all <u>descendent</u> nodes having class containing snippet . Notice that there is a <u>space separating the 2 classes</u>

Can mix with the other
selectors discussed before

CASE SENSITIVITY

CSS Selector	Case Sensitive?
DIV Div DiV	These are equivalent. Tag names are not case sensitive
.article .Article	These are not equivalent. Classes are case sensitive
#a0042 #A0042	Likewise, these are not equivalent. Ids are case sensitive
meta[name] meta[Name]	These are equivalent. The attribute names are not case sensitive
meta[name='description'] meta[name='Description']	These are not equivalent. Attribute values are case sensitive (as principles as class and id)

CSS SELECTOR REFERENCES

https://www.w3schools.com/cssref/css_selectors.asp

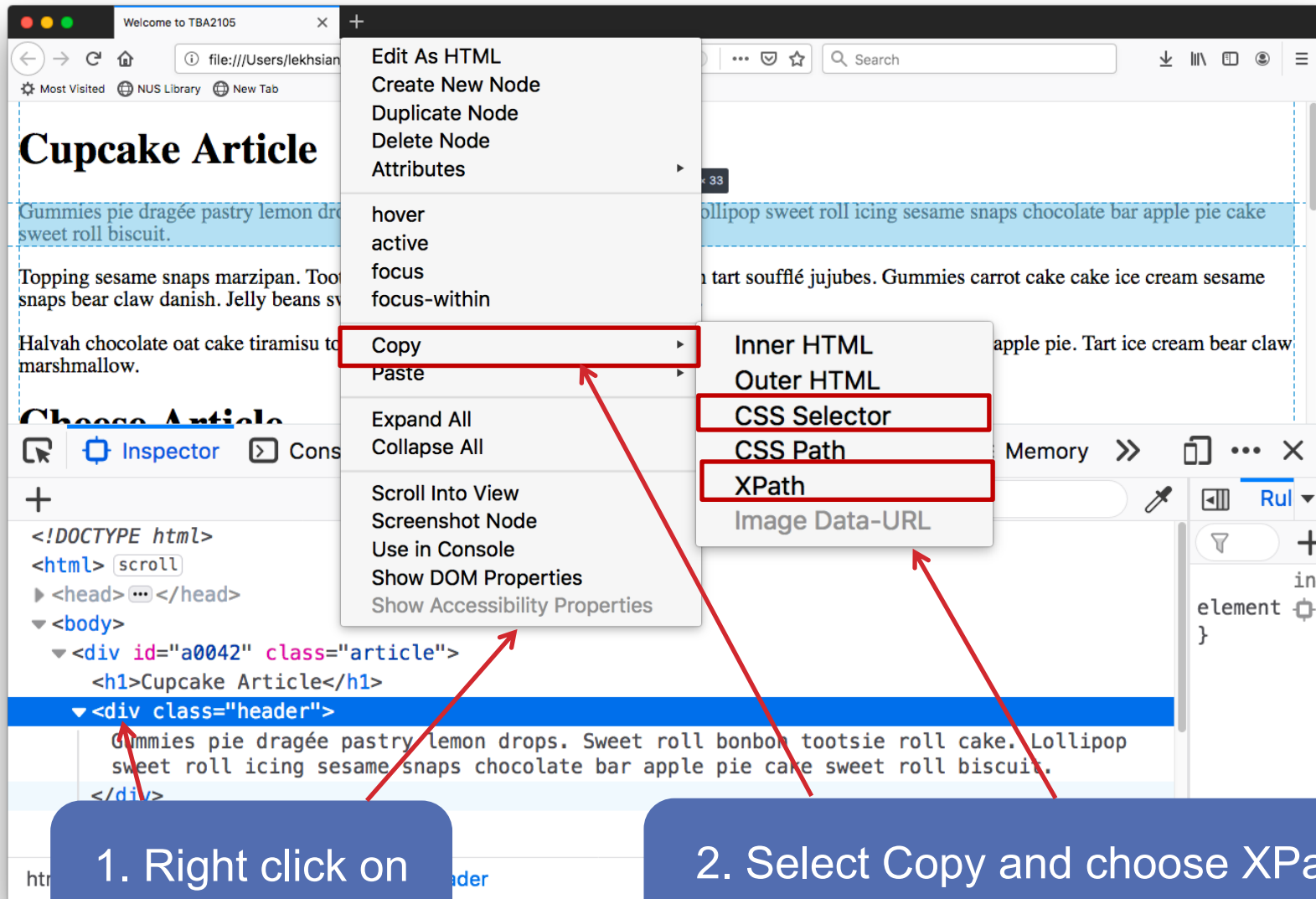
XPath vs CSS Selectors:

<https://johnresig.com/blog/xpath-css-selectors/>

Not confident of writing your own XPath/CSS Selectors?

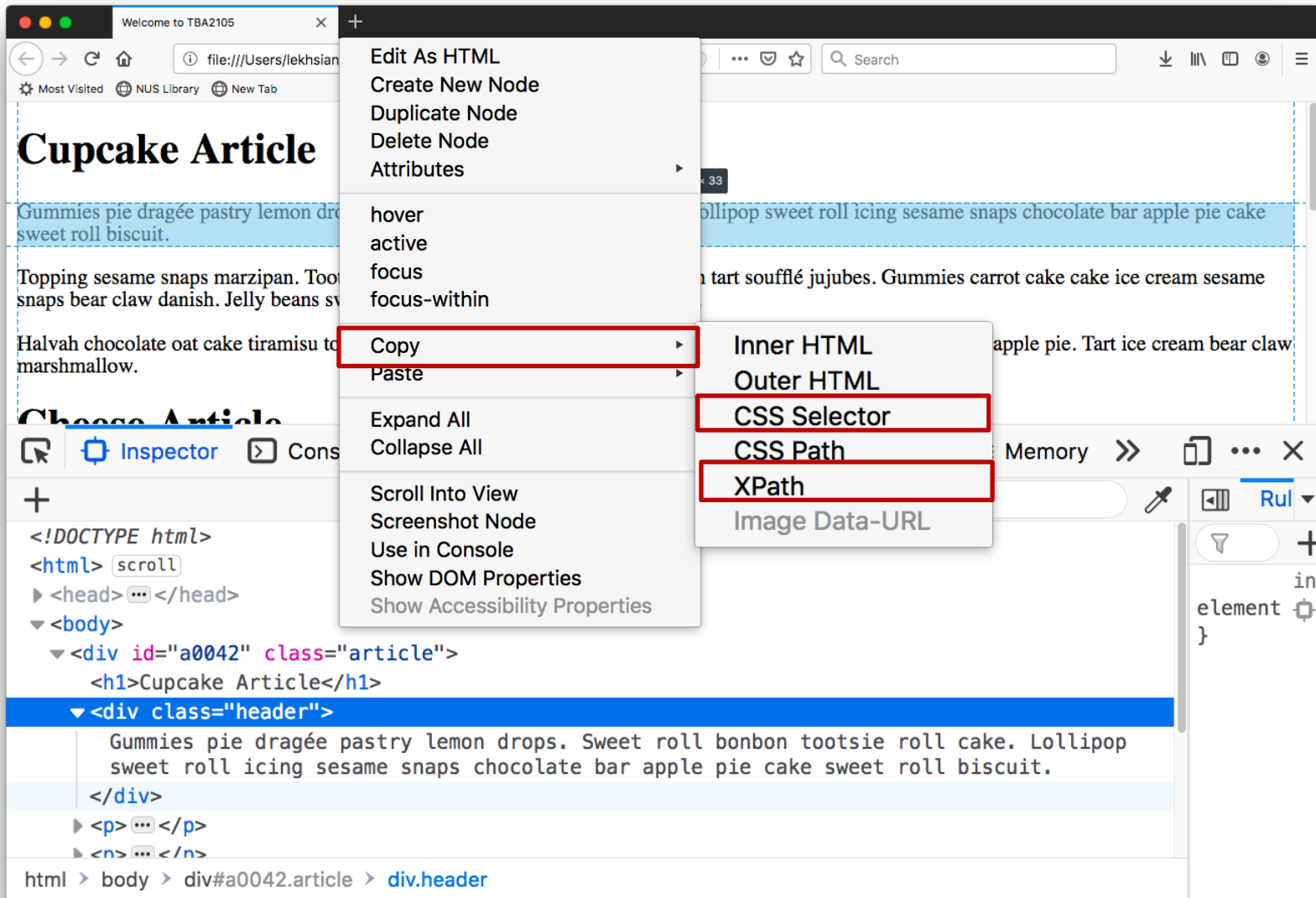
- The browser also allows you to copy the XPath/CSS Selector expressions

COPY XPATH/CSS SELECTOR



COPY XPA

Note that these expressions tend to be very long and overly specific (still better to write manually)



EXTRACTING CONTENT USING HTML PARSER

Document
Object Model
(DOM)

XPath

CSS
Selectors

Extracting
Content using
HTML Parser

HTML PARSING

Different programming platform has its own library for doing **HTML parsing**

- After parsing, we are able to select the HTML elements using XPath and/or CSS selectors
- Can use the **lxml** (for XPath) and **BeautifulSoup** packages

Idea:

- Parse the HTML document
- Use XPath/CSS Selector to select the element(s)
- Extract the value (attribute/text/html) of the element(s)

HANDS-ON: WEB SCRAPING

Download and access:
[Webscrapping using HTML parsing.ipynb](#)

```
jupyter Webscrapping using HTML parsing (autosaved)  Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3

[Icons] Run [Icons] Markdown

XPath approach

In [1]: from lxml import html
import requests

page = requests.get("https://www.comp.nus.edu.sg/~lekhsian/sws3023/page.html")
page.content

Out[1]: b'<!DOCTYPE html>\n<html>\n  <head>\n    <meta charset="UTF-8" />\n    <meta\n      name="description"\n      content="This is a dummy page for learning selectors"\n    />\n    <meta\n      http-equiv="X-UA-Compatible" content="ie=edge" />\n    <title>Welcome to TBA2105</title>\n  </head>\n  <body>\n    <p>this is a paragraph outside any div</p>\n    <div class="article"\n      id="a0042">\n      <h1>Cupcake Article</h1>\n      <div class="header">\n        Gummies pi\n        e drag\&#x3\&#xa9e pastry lemon drops. Sweet roll bonbon tootsie roll\n        cake. Lollipop\n        sweet roll icing sesame snaps chocolate bar apple pie\n        cake sweet roll biscuit.\n      </div>\n      <div>this is a third level div</div>\n    </div>\n    <p>\n      Topping sesame snaps\n      marzipan. Tootsie roll chocolate bar sesame snaps\n      muffin tart souffl\&#x3\&#xa9 jujub\n      es. Gummies carrot cake cake ice cream sesame\n      snaps bear claw danish. Jelly beans\n      sweet roll jujubes caramels cupcake\n      biscuit.\n    </p>\n    <p>\n      Halva\n      h chocolate oat cake tiramisu topping apple pie lollipop jelly-o\n      cake. Topping cot\n      ton candy sweet marzipan apple pie. Tart ice cream bear\n      claw marshmallow.\n    </p>\n    </div>\n    <div class="article" id="a0043">\n      <h1>Cheese Article</h1>\n      <div class="header">\n        The big cheese red leicester rubber cheese. Stilton taleggio\n        halloumi\n        croque monsieur bocconcini cheese triangles cheesecake boursin. Ricotta\n        rancid cheese\n        cheese slices\n        emmental\n        airedale\n        manchego\n        babybel\n        Emmental\n      </div>\n    </div>\n  </body>\n</html>
```

SUMMARY

Document Object Model (DOM)

Navigating the DOM tree

- XPath and CSS Selectors

Extracting Content using HTML Parser

- lxml & BeautifulSoup packages

WHAT'S NEXT?

Mining Web Content III